 

# ASPEN PITKIN COUNTY AP, CO, USA

WMO: **724676**

Lat: **39.230N** Lon: **106.871W** Elev: **2353** StdP: **76.07** Time Zone: **-7.00 (NAM)** Period: **98-19** WBAN: **93073**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-20.1** | **-17.4** | **-23.3** | **0.6** | **-18.9** | **-21.3** | **0.7** | **-15.1** | **8.1** | **-1.0** | **6.9** | **-0.8** | **2.2** | **200** | **0.635** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **17.0** | **29.0** | **12.3** | **27.8** | **12.0** | **26.6** | **11.8** | **14.6** | **23.3** | **13.9** | **22.7** | **13.3** | **22.0** | **4.0** | **340** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **12.4** | **12.0** | **15.6** | **11.4** | **11.2** | **15.3** | **10.3** | **10.4** | **15.3** | **48.4** | **23.4** | **46.5** | **22.6** | **44.6** | **21.9** | **18.0** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **8.2** | **7.1** | **5.8** | DB | **-24.5** | **31.0** | **2.7** | **1.2** | **-26.5** | **31.9** | **-28.1** | **32.6** | **-29.6** | **33.3** | **-31.6** | **34.2** | (4) |
| (5) | | | | WB | **-24.8** | **16.1** | **2.7** | **0.9** | **-26.7** | **16.7** | **-28.3** | **17.2** | **-29.8** | **17.7** | **-31.7** | **18.4** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **5.7** | **-6.1** | **-4.4** | **0.5** | **4.8** | **9.3** | **15.0** | **18.3** | **16.8** | **13.1** | **6.4** | **-0.1** | **-6.1** | (6) |
| (7) | | DBStd | | **9.43** | **4.50** | **4.61** | **4.52** | **4.05** | **4.03** | **3.36** | **2.00** | **2.01** | **3.34** | **4.16** | **5.13** | **4.88** | (7) |
| (8) | | HDD10.0 | | **2354** | **499** | **403** | **293** | **159** | **60** | **5** | **0** | **0** | **12** | **122** | **302** | **498** | (8) |
| (9) | | HDD18.3 | | **4657** | **757** | **636** | **552** | **407** | **279** | **107** | **25** | **56** | **159** | **371** | **552** | **756** | (9) |
| (10) | | CDD10.0 | | **778** | **0** | **0** | **0** | **3** | **39** | **154** | **258** | **210** | **104** | **10** | **0** | **0** | (10) |
| (11) | | CDD18.3 | | **38** | **0** | **0** | **0** | **0** | **0** | **6** | **24** | **7** | **1** | **0** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **1299** | **0** | **0** | **0** | **0** | **19** | **302** | **599** | **300** | **78** | **1** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **252** | **0** | **0** | **0** | **0** | **2** | **52** | **147** | **43** | **8** | **0** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **2.7** | **2.1** | **2.4** | **2.7** | **3.1** | **3.1** | **3.2** | **2.9** | **2.8** | **2.9** | **2.6** | **2.3** | **2.1** | (14) |
| (15) | Precipitation | PrecAvg | | **600** | **54** | **57** | **53** | **64** | **49** | **26** | **53** | **52** | **54** | **47** | **48** | **58** | (15) |
| (16) | | PrecMax | | **733** | **111** | **118** | **115** | **120** | **130** | **51** | **93** | **86** | **105** | **93** | **70** | **117** | (16) |
| (17) | | PrecMin | | **462** | **25** | **26** | **19** | **35** | **14** | **4** | **13** | **19** | **22** | **20** | **10** | **10** | (17) |
| (18) | | PrecStd | | **63** | **24** | **25** | **20** | **20** | **26** | **14** | **20** | **17** | **19** | **20** | **15** | **27** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **7.7** | **9.9** | **17.3** | **20.0** | **26.1** | **30.1** | **31.3** | **29.2** | **27.8** | **22.2** | **16.9** | **9.3** | (19) |
| (20) | | | MCWB | | **1.0** | **2.1** | **5.2** | **6.6** | **10.5** | **12.0** | **13.3** | **12.3** | **12.0** | **8.2** | **5.1** | **1.5** | (20) |
| (21) | | 2% | DB | | **5.0** | **7.4** | **14.0** | **17.6** | **22.7** | **28.0** | **29.6** | **27.8** | **25.5** | **20.0** | **14.0** | **5.9** | (21) |
| (22) | | | MCWB | | **-0.3** | **0.6** | **3.7** | **5.3** | **8.7** | **11.3** | **12.7** | **12.0** | **10.8** | **7.4** | **4.0** | **-0.1** | (22) |
| (23) | | 5% | DB | | **2.8** | **5.4** | **11.3** | **16.2** | **21.1** | **26.8** | **28.3** | **26.8** | **23.8** | **17.8** | **11.4** | **3.2** | (23) |
| (24) | | | MCWB | | **-1.2** | **0.0** | **2.7** | **4.7** | **8.0** | **10.8** | **12.4** | **11.9** | **9.8** | **6.4** | **3.1** | **-1.1** | (24) |
| (25) | | 10% | DB | | **1.2** | **3.1** | **8.8** | **14.0** | **19.0** | **25.1** | **27.1** | **25.1** | **22.2** | **15.9** | **8.5** | **1.3** | (25) |
| (26) | | | MCWB | | **-1.8** | **-1.1** | **1.6** | **4.0** | **7.2** | **10.1** | **12.5** | **11.8** | **9.5** | **5.8** | **1.8** | **-1.7** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **2.2** | **2.9** | **5.6** | **7.4** | **11.6** | **13.9** | **15.7** | **15.1** | **13.5** | **10.0** | **5.8** | **2.3** | (27) |
| (28) | | | MCDB | | **5.3** | **8.0** | **15.8** | **17.7** | **24.1** | **25.6** | **24.0** | **22.9** | **22.6** | **17.0** | **14.6** | **7.2** | (28) |
| (29) | | 2% | WB | | **0.7** | **1.4** | **4.2** | **6.1** | **9.8** | **12.7** | **14.9** | **14.2** | **12.4** | **8.5** | **4.5** | **0.8** | (29) |
| (30) | | | MCDB | | **3.7** | **6.0** | **12.5** | **15.5** | **20.4** | **24.2** | **23.7** | **22.2** | **20.7** | **17.0** | **12.4** | **4.0** | (30) |
| (31) | | 5% | WB | | **-0.5** | **0.3** | **3.0** | **5.3** | **8.7** | **11.8** | **14.2** | **13.7** | **11.5** | **7.4** | **3.4** | **-0.4** | (31) |
| (32) | | | MCDB | | **2.3** | **4.4** | **10.0** | **14.0** | **19.0** | **23.5** | **23.3** | **21.5** | **19.7** | **15.8** | **10.4** | **2.9** | (32) |
| (33) | | 10% | WB | | **-1.6** | **-0.6** | **1.9** | **4.5** | **7.8** | **11.0** | **13.7** | **13.0** | **10.6** | **6.4** | **2.2** | **-1.6** | (33) |
| (34) | | | MCDB | | **1.0** | **3.0** | **7.8** | **12.9** | **17.5** | **22.6** | **22.7** | **20.8** | **18.7** | **14.2** | **7.9** | **1.4** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **12.5** | **12.4** | **12.9** | **13.4** | **14.8** | **17.5** | **17.0** | **16.1** | **16.3** | **15.0** | **13.5** | **12.1** | (35) |
| (36) | | 5% DB | MCDBR | | **14.0** | **14.3** | **16.2** | **17.4** | **18.0** | **19.4** | **19.1** | **18.6** | **18.8** | **18.8** | **17.5** | **13.6** | (36) |
| (37) | | | MCWBR | | **9.8** | **9.1** | **8.3** | **8.1** | **7.7** | **7.7** | **6.9** | **6.8** | **7.5** | **9.1** | **9.4** | **9.0** | (37) |
| (38) | | 5% WB | MCDBR | | **12.0** | **12.6** | **14.8** | **15.7** | **16.8** | **17.9** | **16.1** | **14.8** | **15.1** | **16.3** | **16.2** | **12.0** | (38) |
| (39) | | | MCWBR | | **8.4** | **8.1** | **7.8** | **7.7** | **7.5** | **7.3** | **6.2** | **5.9** | **6.3** | **8.0** | **8.9** | **8.0** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.222** | **0.229** | **0.257** | **0.290** | **0.299** | **0.300** | **0.331** | **0.333** | **0.287** | **0.267** | **0.236** | **0.223** | (40) |
| (41) | | taud | | | **2.438** | **2.404** | **2.393** | **2.422** | **2.512** | **2.522** | **2.484** | **2.451** | **2.603** | **2.617** | **2.585** | **2.475** | (41) |
| (42) | | Ebn at Noon | | | **983** | **1017** | **1010** | **988** | **978** | **973** | **940** | **931** | **966** | **961** | **963** | **958** | (42) |
| (43) | | Edh at Noon | | | **78** | **94** | **107** | **112** | **105** | **104** | **107** | **108** | **87** | **76** | **68** | **70** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **2.62** | **3.50** | **4.95** | **5.78** | **6.33** | **7.48** | **6.64** | **5.82** | **5.20** | **3.84** | **2.73** | **2.18** | (44) |
| (45) | | RadStd | | | **0.14** | **0.22** | **0.39** | **0.39** | **0.49** | **0.47** | **0.36** | **0.29** | **0.34** | **0.32** | **0.24** | **0.18** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (29 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page